Workshop Proposal: Efficient Memory for Long-Context LLM Inference

NeurIPS 2026 Workshop

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1 Title and Workshop Description

Title: *Efficient Memory for Long-Context LLM Inference: Compression, Streaming, and Theoretically-Grounded Methods*

Tagline (140 chars): *Where exact guarantees meet practical streaming: online KV-cache compression, low-rank methods, and the next 128K-token frontier.*

Topic and Content

Autoregressive transformer inference at 32K–128K+ token contexts is bottlenecked by the KV cache, which grows linearly in context length and dominates memory for deployed models. A 7B-parameter model requires ≈4.7 GB for the KV cache alone at T=8,192 tokens in float32—an amount that scales linearly with context. As production serving shifts toward much longer contexts, KV cache management becomes the binding constraint on batch size, GPU utilization, and serving cost.

The space of solutions has fragmented rapidly. *Offline/batch* methods (post-sequence SVD, cross-layer factorizations, quantization-aware low-rank decomposition) are theoretically well-understood but incompatible with streaming generation. *Online/streaming* methods (Oja's rule, eviction policies, sliding windows, attention sinks) are deployment-compatible but often lack formal guarantees or achieve compression only at significant quality cost. *Hardware-aware* approaches (FlashAttention variants, paged attention, speculative decoding with cache reuse) interact with compression in underexplored ways.

A critical gap is the absence of a venue where theoreticians who study low-rank matrix approximation, practitioners deploying at scale, and systems researchers building inference kernels can exchange ideas in real time. Recent arXiv activity (OjaKV, SVDq, xKV, KVTC, KQ-SVD, StreamingLLM, SnapKV, PyramidKV, MagicPIG) reflects a field that is generating results faster than the community can synthesize them.

Why now? Three developments make 2026 the right moment: (i) context windows of 128K–1M tokens are now standard in frontier models, pushing KV memory from a convenience issue to a hard engineering constraint; (ii) a first wave of principled streaming methods (combining randomized linear algebra with online updates) has just appeared, and early cross-paper comparison shows significant quality variation with no agreed-upon benchmark protocol; (iii) hardware vendors are shipping memory-bandwidth-optimized silicon whose impact on compression trade-offs is not yet understood. Without a coordinating venue, the field risks parallel rediscovery, incompatible benchmarking conventions, and missed connections between theory and practice.

Scope and Audiences

The workshop targets four overlapping communities:

- **Theory/algorithms:** randomized linear algebra, streaming algorithms, online learning, matrix sketching.
- **ML systems:** inference engines (vLLM, TensorRT-LLM, TGI), XLA/CUDA kernel authors.
- **LLM practitioners:** fine-tuning, RLHF, long-document QA, multi-turn dialogue, RAG pipelines.

- **Hardware:** accelerator architects interested in memory-hierarchy trade-offs.

Problems we aim to advance:

1. Establish shared benchmarking protocols for online KV compression (reconstruction error, end-task perplexity, latency, memory footprint).
2. Identify which theoretical guarantees (EYM optimality, Halko error bounds, Oja convergence) translate meaningfully to downstream task quality.
3. Surface open questions at the theory-systems interface: XLA-compilable streaming updates, paged-attention integration, rank adaptation across layers.

Concurrent Submissions

The lead organizer has not submitted concurrently related proposals to other venues.

Location Preference

1. Sydney
2. Atlanta
3. Paris

Invited Speakers

Organizers will confirm definitive speaker commitments before the proposal deadline. The list below reflects initial outreach; confirmed speakers are marked **(C)**, tentative **(T)**.

Speaker	Affiliation	Status
Speaker A: online KV / streaming algorithms	TBD	(T)
Speaker B: randomized linear algebra	TBD	(T)
Speaker C: ML systems / inference serving	TBD	(T)
Speaker D: long-context LLM benchmarking	TBD	(T)
Speaker E: industry practitioner	TBD	(T)

We target 4–5 invited speakers with at most 25-minute slots to leave ample time for contributed talks, a poster session, and open discussion.

Diversity and Inclusion

The organizing committee will include researchers spanning career stage (junior PhD students through senior faculty/industry researchers), gender, geographical region (North America, Europe, Asia-Pacific), and sector (academia and industry). We will:

- Issue an explicit diversity statement in the call for papers.
- Actively recruit invited speakers from underrepresented groups and regions.
- Apply for NeurIPS social-good travel grants to subsidize attendance for researchers from under-resourced institutions.
- Use double-blind review for contributed papers.

Estimated Attendance

We estimate 150–300 attendees, based on the intersection of the NeurIPS efficient-inference, long-context LLM, and randomized algorithms communities. No previous iteration of this specific workshop has been held.

Special Requirements

A poster session area for 20–30 posters is requested. No unusual A/V or compute requirements beyond standard presentation setup.

Differentiation from Related Workshops

Recent NeurIPS/ICML workshops on efficient transformers (e.g., EfficientML, sparsity-focused workshops) have emphasized pruning, quantization, and distillation. KV cache compression, particularly streaming/online methods with theoretical guarantees, has not been the primary focus of any recent workshop. Our workshop is distinguished by: (i) exclusive focus on memory—not speed, sparsity, or model compression broadly; (ii) explicit emphasis on bridging theory (randomized NLA, online learning) with practice (inference engines, benchmarks); (iii) a live benchmarking session where submitted methods are evaluated on a shared held-out protocol announced at submission time.

Preliminary Benchmark Findings (Pilot for the Live Benchmark Session)

To validate that the shared protocol in item (1) above is measurable and discriminative, the organizers ran a pilot sweep on a single commodity workstation using GPT-2-small key/value activations extracted from WikiText-2, comparing exact streaming SVD (Brand rank-1 update, "TKV"), Oja's-rule online compression ("OjaKV"), and a frozen low-rank basis. The pilot exercises every axis of the proposed protocol—reconstruction error, memory footprint, end-task perplexity, and per-token latency—and confirms that these axes separate methods cleanly:

- **Reconstruction error (real activations).** Averaged over 15 layer/head configurations, exact streaming SVD reduced relative reconstruction error by 2.0×–6.6× over OjaKV at matched rank (e.g. at T=512, rank 32: 0.23 vs. 0.97). The EYM-optimality guarantee translates directly into measured quality.
- **Memory at scale.** At T=128K tokens the full float32 KV cache requires 9.44 GB; a frozen-pyramid basis compresses this to 2.21 GB (4.3×), and beat OjaKV on reconstruction error at every long-context setting tested (1K–32K).
- **End-task perplexity.** A frozen low-rank basis (average rank 15) achieved perplexity 89.6 versus 129.6 for the exact cache (ratio 0.69), indicating that a well-chosen fixed basis can match—and here, regularize below—full-cache quality on this evaluation.
- **Cheap quality levers.** Importance-weighted cold-start cut reconstruction error by 90–95% at rank ≥ 8, and a full-precision sink+window buffer drove sink-token reconstruction error to exactly 0 at a cost of < 0.1 MB per head.

These numbers are a single-workstation pilot, not a leaderboard; their purpose is to demonstrate that the live-benchmark session can rank submitted methods on a shared, reproducible protocol. Full scripts and figures accompany this proposal.

Workshop Schedule (Proposed)

Time	Session
08:30–09:00	Opening remarks and problem framing
09:00–11:00	Invited talks (2 × 25 min + discussion)
11:00–12:00	Contributed talks (3 × 15 min)
12:00–13:30	Lunch + Poster session
13:30–15:00	Invited talks (2 × 25 min + discussion)
15:00–16:00	Live benchmark reveal and community discussion
16:00–17:00	Panel: "What does theory owe practice, and vice versa?"
17:00–17:30	Contributed talks (2 × 12 min)
17:30–18:00	Open discussion and closing

Contributed work will be solicited via OpenReview (required per NeurIPS guidelines). All accepted submissions will be non-archival; authors may optionally post on arXiv. Accept/reject notification will be issued by **September 29, 2026** at the latest.

Organizer Information

Organizing Committee

David Aror `tejirid.aror@gmail.com` Independent Research

David Aror is an independent researcher working on efficient inference for large language models, with a current focus on online KV cache compression using exact streaming SVD updates (Brand's rank-1 update) and randomized linear algebra (Halko randomized range finder). His recent work, TKV, is the first online KV compression method with a per-token Eckart-Young-Mirsky optimality guarantee. He has experience running small research communities and organizing reading groups on efficient ML. He is not submitting any other workshop proposals to NeurIPS 2026.

Xavistin C. Arul Arasu `js01.christopher@gmail.com` Independent Research

Xavistin C. Arul Arasu is an independent researcher working on efficient inference systems for large language models, with a focus on streaming KV-cache compression and the empirical evaluation of low-rank and sink/window hybrid methods. He led the benchmarking effort behind this proposal—reconstruction-error, memory-ratio, perplexity, and streaming-error sweeps comparing exact streaming SVD (Brand update) against Oja's-rule baselines on GPT-2—which forms the basis for the shared live-benchmark protocol described in Section 1. He is not submitting any other workshop proposals to NeurIPS 2026.

Program Committee

The PC will be recruited to ensure each submission receives at least 3 reviews and no reviewer is assigned more than 3 papers. We estimate 60–100 submissions (based on submission rates at adjacent workshops) and therefore need 60–100 reviewers.

Confirmed PC members (to be updated before submission):

- PC Member 1, Affiliation **(C)**
- PC Member 2, Affiliation **(C)**
- . . . (target: 60+ members to cover expected submission volume)

Organizers will recuse themselves from reviewing any submission with which they have a conflict of interest as defined by the NeurIPS CoI policy. Specifically, no organizer will be involved in assessing submissions from their own organization. Conflict detection will use OpenReview's built-in CoI tooling.

Workshop organizers will not submit papers to this workshop (per NeurIPS policy).

References

[1] Anonymous. OjaKV: Online KV cache compression via Oja's rule. arXiv:2509.21623, 2025.

[2] N. Halko, P.-G. Martinsson, J. A. Tropp. Finding structure with randomness. *SIAM Review*, 53(2):217–288, 2011.

[3] M. Brand. Fast low-rank modifications of the thin SVD. *Linear Algebra and its Applications*, 415(1):20–30, 2006.

[4] G. Xiao et al. Efficient streaming language models with attention sinks. arXiv:2309.17453, 2023.

[5] Anonymous. SVDq. arXiv:2502.15304, 2025.

[6] Anonymous. xKV. arXiv:2503.18893, 2025.